Riemann Integration

Let $f\colon [a,b] \to \mathbb{R}$ be a function. What does it mean to say that f is Riemann integrable? How does one define integral $\int_a^b f(x)\,dx$ (or $\int_a^b f$) ?

Definition 1 : Let $a, b \in \mathbb{R}$ with $a < b$. Consider the closed and bounded interval $[a,b]$. A finite subset $P = \{x_0, x_1, x_2, \ldots, x_n\} \subseteq [a,b]$ is said to be a *partition* of $[a,b]$ if $a = x_0 < x_1 < x_2 < \cdots < x_n = b$ for some $n \geq 1$. Thus a partition P of $[a,b]$ subdivide interval $[a,b]$ into non-overlapping subintervals

$$I_1 = [x_0, x_1],\ I_2 = [x_1, x_2],\ \ \ldots,\ I_n = [x_{n-1}, x_n].$$

By *mesh* (or *norm*) of the partition P, we mean the length of the largest subinterval $[x_{i-1}, x_i]$ for $1 \leq i \leq n$. We set $\Delta x_i = x_i - x_{i-1}$ $(1 \leq i \leq n)$. Thus the mesh of P, denoted by $\mu(P)$, is given by

$$\mu(P) = \max\{\Delta x_i : 1 \leq i \leq n\}.$$

A partition P' of $[a,b]$ is said to be a *refinement* of P if $P \subseteq P'$. Clearly, in this case, $\mu(P') \leq \mu(P)$. If P_1 and P_2 are partitions of $[a,b]$, then $P_1 \cup P_2$ is a refinement of P_1 as well as P_2, and it is called the *common refinement* of P_1 and P_2. The set of all partitions of $[a,b]$ is denoted by $\mathcal{P}ar[a,b]$.

Given a partition $P = \{x_0, x_1, x_2, \ldots, x_n\}$ of $[a,b]$, a collection $T = \{t_1, t_2, \ldots, t_n\}$, where $t_i \in [x_{i-1}, x_i]$ for $1 \leq i \leq n$, is called a *tag* for the partition P. Sometimes P together with a choice of tag T is called a *tagged partition* (P,T) of $[a,b]$.

Let $f\colon [a,b] \to \mathbb{R}$ be a function. Then for any tagged partition (P,T) of $[a,b]$, we set

$$R(f,P,T) = \sum_{i=1}^{n} f(t_i)\Delta x_i\ ,$$

where $P = \{x_0, x_1, x_2, \ldots, x_n\}$ and $t_i \in [x_{i-1}, x_i] \quad (1 \leq i \leq n)$. The quantity $R(f,P,T)$ is called the *Riemann sum* of f associated with a tagged partition (P,T) of $[a,b]$.

Definition 2 : (Riemann Integrability) Let $f\colon [a,b] \to \mathbb{R}$ be a function. A real number L is said to be a *Riemann integral* of f over $[a,b]$ if $\forall\ \epsilon > 0,\ \exists\ \delta > 0$ ($\delta = \delta(\epsilon)$ depends on ϵ and f) such that for any partition P of $[a,b]$ with mesh $\mu(P) < \delta$, we have

$$|R(f,P,T) - L| < \epsilon,$$

for every choice of tag T for the partition P. In this case, we write

$$L = \int_a^b f(x)\,dx = \lim_{\mu(P)\to 0} R(f,P,T).$$

and we say that f is Riemann integrable over $[a,b]$. The set of all integrable functions over $[a,b]$ is denoted by $\mathcal{R}[a,b]$.

Observations/Examples :

1. In Definitions 1 and 2, we could take $a=b$. Then $[a,b]=\{a\}$ is a singleton set and every real-function f on $\{a\}$ is integrable with $\int_a^a f(x)\,dx = 0$.

2. If $f \in \mathcal{R}[a,b]$, then Riemann integral L of f over $[a,b]$ is unique. (Solution : Suppose L and L' are both Riemann integrals of f over $[a,b]$. Then for given $\epsilon > 0$, there is a $\delta > 0$ such that for every partition $P \in \mathcal{P}ar[a,b]$ with $\mu(P) < \delta$, we have $|R(f,P,T) - L| < \frac{\epsilon}{2}$ and $|R(f,P,T) - L'| < \frac{\epsilon}{2}$. Clearly,

   $$|L - L'| = |L - R(f,P,T) + R(f,P,T) - L'| \leq |L - R(f,P,T)| + |R(f,P,T) - L'| < \epsilon.$$

   As $\epsilon > 0$ is arbitrary, we have $L = L'$.)

3. If $f \in \mathcal{R}[a,b]$, then f must be bounded. (Solution : Suppose L is Riemann integral of f. Then for $\epsilon = 1$, there exists a $\delta > 0$ such that for a tagged partition (P,T) with $\mu(P) < \delta$, we have $|R(f,P,T) - L| < 1$. Let $P = \{x_0, x_1, \ldots, x_n\}$ and a collection $T = \{t_1, t_2, \ldots, t_n\}$ be a tag for P. Suppose, if possible, f is not bounded over $[a,b]$. Then f is unbounded over at least one subinterval, say $[x_{i_0-1}, x_{i_0}]$. Now choose a point, $t'_{i_0} \in [x_{i_0-1}, x_{i_0}]$ such that $|f(t'_{i_0}) - f(t_{i_0})|\Delta x_{i_0} > 2$. Consider another tag $T_0 = (T \setminus \{t_{i_0}\}) \cup \{t'_{i_0}\} = \{t_1, \ldots, t_{i_0-1}, t'_{i_0}, t_{i_0+1}, \ldots, t_n\}$ for P. We see that $|R(f,P,T') - R(f,P,T)| = |f(t'_{i_0}) - f(t_{i_0})|\Delta x_{i_0} > 2$. This is a contradiction as

   $$|R(f,P,T') - R(f,P,T)| \leq |R(f,P,T') - L| + |L - R(f,P,T)| < 2.$$

   Therefore, f must be bounded over $[a,b]$.)

4. If $f(x) = p$ is a constant function, then $f \in \mathcal{R}[a,b]$ and $\int_a^b f(x)\,dx = p\ (b-a)$. (Solution : For every tagged partition (P,T) of $[a,b]$, we see that the Riemann sum $R(f,P,T) = \sum_{i=1}^n\ f(t_i)\Delta x_i = \sum_{i=1}^n p\Delta x_i = p\sum_{i=1}^n \Delta x_i = p(b-a)$.)

5. If $f \in \mathcal{R}[a,b]$ and $f \geq 0$, then $\int_a^b f\,dx \geq 0$. More generally, if $f, g \in \mathcal{R}[a,b]$ and $f \leq g$, then $\int_a^b f\,dx \leq \int_a^b g\,dx$.

Theorem 1 : (Linearity of Riemann integrals) Let $f, g \in \mathcal{R}[a,b]$ and $c_1, c_2 \in \mathbb{R}$. Then $c_1 f + c_2 g$ defined by $(c_1 f + c_2 g)(x) = c_1(f(x)) + c_2(g(x))$ for $x \in [a,b]$, is Riemann integrable over $[a,b]$ and

$$\int_a^b (c_1 f + c_2 g)(x)\ \ dx = c_1\ \int_a^b f(x)\,dx + c_2\ \int_a^b g(x)\,dx\ .$$

(Proof : As $f, g \in \mathcal{R}[a,b]$, for every $\epsilon > 0$, there is $\delta > 0$ such that for every tagged partition (P,T) of $[a,b]$ with $\mu(P) < \delta$, we have

$$\left| R(f,P,T) - \int_a^b f\, dx \right| < \frac{\epsilon}{2(|c_1|+1)} \quad \text{and} \quad \left| R(g,P,T) - \int_a^b g\, dx \right| < \frac{\epsilon}{2(|c_2|+1)} \ .$$

Also, we have $R(c_1 f + c_2 g, P, T) = c_1\ R(f,P,T) + c_2\ R(g,P,T)$. Now clearly,

$$\left| R((c_1 f + c_2 g), P, T) - (c_1 \ \int_a^b f\, dx + c_2 \ \int_a^b g\, dx) \right| < \epsilon \ .$$

This completes the proof.)

Definition 3 : (Darboux Integrability) Let $f\colon [a,b] \to \mathbb{R}$ be a bounded function and $|f(x)| \le M,\ \forall x \in [a,b]$. For a partition $P = \{x_0, x_1, \ldots, x_n\}$ of $[a,b]$ and $1 \le i \le n$, let

$$m_i(f) = m_i = \inf_{t \in [x_{i-1}, x_i]} f(t) \quad \text{and} \quad M_i(f) = M_i = \sup_{t \in [x_{i-1}, x_i]} f(t) \ .$$

Then $L(f,P) = \sum_{i=1}^n m_i\ \Delta x_i$ and $U(f,P) = \sum_{i=1}^n M_i\ \Delta x_i$ are called *lower sum* and *upper sum* of f with respect to partition P, respectively. Clearly, we have

$$-M\ (b-a)\ \le\ L(f,P)\ \le\ U(f,P)\ \le\ M\ (b-a)\ .$$

The *lower* and *upper integrals* of f over $[a,b]$ are respectively given by

$$\underline{\int_a^b} f\, dx = \sup_{P \in \mathcal{P}ar[a,b]} L(f,P) \quad \text{and} \quad \overline{\int_a^b} f\, dx = \inf_{P \in \mathcal{P}ar[a,b]} U(f,P) \ .$$

The bounded function f is said to be *Darboux integrable* over $[a,b]$ if $\underline{\int_a^b} f\, dx = \overline{\int_a^b} f\, dx$ and the common value is called the *Darboux integral* of f over $[a,b]$.

Observations / Examples :

1. If $P \subseteq P'$ are partitions of $[a,b]$, then for a bounded function $f\colon [a,b] \to \mathbb{R}$, we have $L(f,P) \le L(f,P')$ and $U(f,P') = U(f,P)$. (Solution : First suppose that the refinement P' has exactly one extra point. Let $P = \{x_0, x_1, \ldots, x_n\}$ and $P' = \{x_0, \ldots, x_{j-1}, x^*, x_j, \ldots, x_n\}$, where $x_{j-1} < x^* < x_j$. Let

   $$m_j = \inf_{t \in [x_{j-1}, x_j]} f(t),\ w_1 = \inf_{t \in [x_{j-1}, x^*]} f(t) \text{ and } w_2 = \inf_{t \in [x^*, x_j]} f(t).$$

   Clearly, $m_j \le w_1,\ m_j \le w_2$, and $L(f,P') - L(f,P) = w_1(x^* - x_{j-1}) + w_2(x_j - x^*) - m_j(x_j - x_{j-1}) = (w_1 - m_j)(x^* - x_{j-1}) + (w_2 - m_j)(x_j - x^*) \ge 0$. If P' has k extra points, then repeat this argument k times, we get $L(f,P) \le L(f,P')$. Similarly, we show that $U(f,P') \le U(f,P)$ or note that $U(-f,P) = -L(f,P)$ and $L(-f,P) = -U(f,P)$.)

2. (Darboux integrability criterion) A bounded function $f\colon [a,b] \to \mathbb{R}$ is Darboux integrable if and only if for every $\epsilon > 0$, there is a partition $P \in \mathcal{P}ar[a,b]$ such that $U(f,P) - L(f,P) < \epsilon$. (Solution : Let f is Darboux integrable over $[a,b]$ and $L = \underline{\int}_a^b f = \overline{\int}_a^b f$. Let $\epsilon > 0$ be given. As, $L = \underline{\int}_a^b f = \sup\limits_{P \in \mathcal{P}ar[a,b]} L(f,P)$ and $L = \overline{\int}_a^b f = \inf\limits_{P \in \mathcal{P}ar[a,b]} U(f,P)$, there are partitions $P_1, P_2 \in \mathcal{P}ar[a,b]$ such that $L - \frac{\epsilon}{2} < L(f,P_1)$ and $U(f,P_2) < L + \frac{\epsilon}{2}$. Let $P = P_1 \cup P_2$ be the common refinement. Then
$$L - \frac{\epsilon}{2} < L(f,P_1) \le L(f,P) \le U(f,P) \le U(f,P_2) < L + \frac{\epsilon}{2}.$$
Thus, $U(f,P) - L(f,P) < \epsilon$. Conversely, suppose the given conditions hold. As, $L(f,P) \le \underline{\int}_a^b f \le \overline{\int}_a^b f \le U(f,P)$, we see that $0 \le \overline{\int}_a^b f - \underline{\int}_a^b f < \epsilon$. As $\epsilon > 0$ is arbitrary, we have $\underline{\int}_a^b f = \overline{\int}_a^b f$.)

3. The Dirichlet function f on $[0,1]$ is not Darboux integrable. (Solution : For any partition $P = \{x_0, x_1, \ldots, x_n\}$ of $[0,1]$, we see that $m_i = 0$ and $M_i = 1$. Thus $L(f,P) = 0$ and $U(f,P) = 1$. Hence, $0 = \underline{\int}_a^b f \neq \overline{\int}_a^b f = 1$.)

Theorem 2 : Let $f\colon [a,b] \to \mathbb{R}$ be a bounded function. Then f is Riemann integrable if and only if f is Darboux integrable. (Proof : For any tagged partion (P,T) of $[a,b]$, we have $L(f,P) \le R(f,P,T) \le U(f,P)$. Let $f \in \mathcal{R}[a,b]$ and $L = \int_a^b f$. For given $\epsilon > 0$, there is a $\delta > 0$ such that for any $P \in \mathcal{P}ar[a,b]$ with $\mu(P) < \delta$, we have $|R(f,P,T) - L| < \frac{\epsilon}{4}$ for any choice of tag T for P. Fix a partition $P_0 = \{x_0, x_1, \ldots, x_n\}$ with $\mu(P_0) < \delta$. Now choose tags $T = \{t_1, \ldots, t_n\}$ and $T' = \{t'_1, \ldots, T'_n\}$ so that $f(t_i)$ is very close to m_i and $f(t'_i)$ is very close to M_i. In fact, we can ensure that
$$R(f,P_0,T) - L(f,P_0) < \frac{\epsilon}{4} \quad \text{and} \quad U(f,P_0) - R(f,P_0,T') < \frac{\epsilon}{4}.$$
Also, $|R(f,P_0,T') - R(f,P_0,T)| \le |R(f,P_0,T') - L| + |L - R(f,P_0,T)| < \frac{\epsilon}{2}$. Thus, we see that $U(f,P_0) - L(f,P_0)$
$$\begin{aligned} &\le U(f,P_0) - R(f,P_0,T') + R(f,P_0,T') - R(f,P_0,T) + R(f,P_0,T) - L(f,P_0) \\ &< \frac{\epsilon}{4} + \frac{\epsilon}{2} + \frac{\epsilon}{4} = \epsilon. \end{aligned}$$
Hence, f is Darboux integrable. Conversely, suppose f is Darboux integrable over $[a,b]$. Let $\epsilon > 0$ be given. Then there is a partition, say $P_1 = \{y_0, y_1, \ldots, y_{n_1}\}$ of $[a,b]$ such that $U(f,P_1) - L(f,P_1) < \frac{\epsilon}{2}$ and $L = \underline{\int}_a^b f = \overline{\int}_a^b f$. Let $\delta = \frac{\epsilon}{8Mn_1} > 0$. Consider a partition $P \in \mathcal{P}ar[a,b]$ with $\mu(P) < \delta$. We shall show that $|R(f,P,T) - L| < \epsilon$ for any tag T for P. Let $P^* = P \cup P_1$ be the common refinement. Then by refinement priciple, $L(f,P_1) \le L(f,P^*) \le U(f,P^*) \le U(f,P_1)$ and we have $U(f,P^*) - L(f,P^*) < \frac{\epsilon}{2}$. If

$P = \{x_0, x_1, \ldots, x_n\}$ and $P^* = \{x_0^*, x_1^*, \ldots, x_{n_2}^*\}$, then in $U(f,P) - U(f,P^*)$ identical terms cancel out and there are at most $n_1 - 1$ terms corresponding to x_j^* with $x_{i-1} < x_j^* < x_i$ and each is of magnitude at most $M\delta$. Thus,

$$U(f,P) - U(f,P^*) < (n_1 - 1)2M\delta < \frac{\epsilon}{4}.$$

Similarly, $L(f,P^*) - L(f,P) < \frac{\epsilon}{4}$. Thus,

$$U(f,P) - L(f,P) = U(f,P) - U(f,P^*) + U(f,P^*) - L(f,P^*) + L(f,P^*) - L(f,P) < \epsilon.$$

As $L(f,P) \leq R(f,P,T) \leq U(f,P)$ and $L(f,P) \leq L \leq U(f,P)$, we must have $|R(f,P,T) - L| < \epsilon$. Hence, f is Riemann integrable and $\int_a^b f\,dx = L$.)

We note that $f \in \mathcal{R}[a,b]$ if and only if for every $\epsilon > 0$, there is a partition $P \in \mathcal{P}ar[a,b]$ such that $U(f,P) - L(f,P) < \epsilon$. This is called the *Riemann's integrability criterion.*

Observations / Examples :

1. Every continuous function $f\colon [a,b] \to \mathbb{R}$ is Riemann integrable over $[a,b]$. (Solution : Let $\epsilon > 0$ be given. As $[a,b]$ is compact, f is uniformly continuous on $[a,b]$. Thus, there is a $\delta > 0$ such that whenever $s,t \in [a,b]$ with $|s-t| < \delta$, we have $f(s) - f(t)| < \frac{\epsilon}{(b-a)+1}$. Choose a partition $P = \{x_0, x_1, \ldots, x_n\}$ with mesh $\mu(P) < \delta$. We know that a continuous function on a compact set attains its bounds. Thus there are $u_i, v_i \in [x_{i-1}, x_i]$ such that $M_i(f) = f(u_i)$ and $m_i(f) = f(v_i)$ for $1 \leq i \leq n$. As $|u_i - v_i| < \delta$, we have $M_i(f) - m_i(f) < \frac{\epsilon}{(b-a)+1}$. Hence,

   $$U(f,P) - L(f,P) = \sum_{i=1}^{n} (M_i - m_i)\ \Delta x_i < \sum_{i=1}^{n} \frac{\epsilon}{(b-a)+1}\ \Delta x_i < \epsilon.$$

   By Riemann's integrability criterion, $f \in \mathcal{R}[a,b]$.)

2. Every monotonic function $f\colon [a,b] \to \mathbb{R}$ is Riemann integrable over $[a,b]$. (Solution : Assume that f is monotonically increasing. Otherwise, replace f with $-f$. Let $\epsilon > 0$ be given. Choose n so that $\frac{(b-a)(f(b)-f(a))}{n} < \epsilon$. Consider the partition $P = \{x_0, x_1, \ldots, x_n\}$ such that $\Delta x_i = \frac{b-a}{n}$. In other words, $x_i = a + i\left(\frac{b-a}{n}\right)$ and P divides the interval $[a,b]$ into n subintervals of equal lengths. As f is monotonically increasing, we see that $m_i = f(x_{i-1})$ and $M_i = f(x_i)$ for $1 \leq i \leq n$. Now, $U(f,P) - L(f,P) = \sum_{i=1}^{n}(f(x_i) - f(x_{i-1}))\left(\frac{b-a}{n}\right) = \left(\frac{b-a}{n}\right)\sum_{i=1}^{n}(f(x_i) - f(x_{i-1})) = \left(\frac{b-a}{n}\right)(f(b) - f(a)) < \epsilon$. By Riemann's integrability criterion, $f \in \mathcal{R}[a,b]$.)

3. The Thomae function g on $[0,1]$ is Riemann integrable. (Solution : Let $\epsilon > 0$ be given. Choose $m \in \mathbb{N}$ such that $\frac{1}{m} < \frac{\epsilon}{2}$. Note that there are only finitely many

rational numbers in the standard form $\frac{p}{q} \in [0,1]$ with $1 \leq q < m$. Let N be the total number of such rationals in $[0,1]$. Now consider a partition $P = \{x_0, x_1, \ldots, x_n\}$ of $[0,1]$ such that each of the rational number $\frac{p}{q} \in [0,1]$ with $1 \leq q < m$ is contained in at most one subinterval $I_i = [x_{i-1}, x_i]$ of the partition P. Further, if $\frac{p}{q} \in I_i$ with $1 \leq q < m$, then the length of the subinterval $I_i = \Delta x_i < \frac{\epsilon}{2N}$. Clearly, $L(g,P) = 0$ and $U(g,P) = \sum_{i \in A} M_i(g)\Delta x_i + \sum_{j \in B} M_j(g)\Delta x_i$, where in the first sum the subintervals I_i for $i \in A$ contain exactly one point of the form $\frac{p}{q} \in I_i$ with $1 \leq q < m$, while the subintervals I_j for $j \in B$ in the second sum do not contain any such point. Thus all rational numbers $\frac{a}{b} \in I_j$ for $j \in B$ (in standard form) has $b \geq m$. Here, we have $A \coprod B = \{1, 2, \ldots, n\}$ and $|A| = N$. Now $\sum_{i \in A} M_i(g)\Delta x_i < \sum_{i \in A} 1 \ \frac{\epsilon}{2N} = \frac{\epsilon}{2}$. Also, $M_j(g) \leq \frac{1}{m} < \frac{\epsilon}{2}$ for $j \in B$. Thus $\sum_{j \in B} M_j(g)\Delta x_j < \frac{\epsilon}{2} \sum_{j \in B} \Delta x_j < \frac{\epsilon}{2}$. This shows that $U(g,P) < \epsilon$ and $U(g,P) - L(g,P) < \epsilon$. Hence, by Riemann integrability criterion, $g \in \mathcal{R}[0,1]$.)

Definition 4 : A subset $A \subseteq \mathbb{R}$ is called a *zero set* (or *measure zero set*) if for every $\epsilon > 0$, there is countable family of open intervals $\{I_j = (a_j, b_j) : j \in \Lambda\}$ ($\Lambda \subseteq \mathbb{N}$) such that $A \subseteq \bigcup_{j \in \Lambda} I_j$ and $\sum_{j \in \Lambda}(b_j - a_j) < \epsilon$. In other words, a subset $A \subseteq \mathbb{R}$ is a zero set if for every $\epsilon > 0$, it can be covered by countably many open intervals of total length $< \epsilon$. (Examples : Every finite subsets of reals is a zero set. Every countable subset of reals is also a zero set. A subset of a zero set is a zero set. Every countable union of zero sets is a zero set. It can be shown that the Cantor ternary set is an uncountable zero set.)

Definition 5 : Let (X, d) be a metric space and $f \colon X \to \mathbb{R}$ be a function. For $r > 0$ and $x \in X$, we define

$$\Omega_f(x, r) = \Omega_f(x, B_d(x; r)) = \sup\{|f(u) - f(v)| : u, v \in B_d(x; r)\} \in \mathbb{R} \cup \{\infty\},$$

and $\omega_f(x) = \inf\{\Omega_f(x, r) : r > 0\} \in \mathbb{R} \cup \{\infty\}$. ($\omega_f(x)$ is called the *oscillation* of f at x). Exercise : Show that f is continuous at $x \in X$ if and only if $\omega_f(x) = 0$. (Solution : Let f be continuous at $x \in X$ and $\epsilon > 0$ be given. Then there is $\delta > 0$ such that

$$f(B_d(x;\delta)) \subseteq \left(f(x) - \frac{\epsilon}{2},\ f(x) + \frac{\epsilon}{2}\right).$$

This shows that for every $u, v \in B_d(x;\delta)$, we have $|f(u) - f(v)| < \epsilon$ and $\Omega_f(x,\delta) < \epsilon$. Hence, $\omega_f(x) < \epsilon$. As $\epsilon > 0$ is arbitrary, we have $\omega_f(x) = 0$. Conversely, if $\omega_f(x) = 0$, then for a given $\epsilon > 0$, there is a $\delta > 0$ such that $\Omega_f(x,\delta) < \epsilon$. Hence, we see that $f(B_d(x;\delta)) \subseteq (f(x) - \epsilon,\ f(x) + \epsilon)$. This shows that f is continuous at x.)

Theorem 3 : (Riemann-Lebesgue Theorem) A bounded function $f \colon [a,b] \to \mathbb{R}$ is Riemann integrable if and only if the set $E_f = \{x \in [a,b] : f \text{ is discontinuous at } x\}$ of discontinuities of f is a zero set. In other words, $f \in \mathcal{R}[a,b]$ if and only if f is continuous almost everywhere on $[a,b]$. (Proof: First suppose that $f \in \mathcal{R}[a,b]$. Let $\epsilon > 0$ be given and for each $k \in \mathbb{N}$, set

$D_k = \{x \in [a,b] : \omega_f(x) > \frac{1}{2^k}\}$. We shall show that D_k is contained in a finite union of open intervals of total length $< \frac{\epsilon}{2^k}$. By Riemann's integrability criterion, there is a partition $P_k = \{x_0^k, x_1^k, \ldots, x_{n(k)}^k\}$ such that $U(f,P_k) - L(f,P_k) = \sum_{i=1}^{n(k)}(M_i^k - m_i^k)\Delta x_i < \frac{\epsilon}{4^k}$, where M_i^k and m_i^k are supremum and infimum of f over the subinterval $[x_{i-1}^k, x_i^k]$. If $x \in D_k \cap (x_{i-1}^k, x_i^k)$, then there is $r > 0$ such that $(x-r, x+r) \subseteq (x_{i-1}^k, x_i^k)$ and we have $\frac{1}{2^k} < \omega_f(x) \leq \Omega_f(x,r) \leq M_i^k - m_i^k$. If $\sum'$ denotes summation over i such that $D_k \cap (x_{i-1}^k, x_i^k) \neq \emptyset$, then we have $\frac{1}{2^k}\sum' \Delta x_i \leq U(f,P_k) - L(f,P_k) < \frac{\epsilon}{4^k}$. Thus $\sum' \Delta x_i < \frac{\epsilon}{2^k}$. Since there are only finitely many points of D_k contained in the partition P_k, we concluded that D_k is contained in a union of finitely many open intervals of total length $< \frac{\epsilon}{2^k}$. As $E_f = \bigcup_{k=1}^{\infty} D_k$, we see that E_f is contained in countably many open intervals of total length $< \sum_{k=1}^{\infty} \frac{\epsilon}{2^k} = \epsilon$. Since $\epsilon > 0$ is arbitrary, E_f is a zero set.

Conversely, let f be a function on $[a,b]$ with $|f(x)| \leq M$, $\forall x \in [a,b]$ and the set E_f of discontinuities of f is a zero set. Given $\epsilon > 0$, choose $k \in \mathbb{N}$ such that $\frac{1}{k} < \frac{\epsilon}{2((b-a)+1)}$. Let $D_k' = \{x \in [a,b] : \omega_f(x) \geq \frac{1}{k}\}$. As $D_k' \subseteq E_f$, D_k' is a zero set. Thus $D_k' \subseteq \bigcup_{j=i}^{\infty} J_j$ such that $\sum_{j=1}^{\infty}(b_j - a_j) \leq \frac{\epsilon}{4M}$, where $J_j = (a_j, b_j)$. Also, for $x \in [a,b] \setminus D_k'$, there is an open interval I_x containing x such that $\omega_f(x) \leq \Omega_f(x, I_x) < \frac{1}{k}$. Now, let

$$\mathcal{C} = \{J_j ; j \in \mathbb{N}\} \cup \{I_x : x \in [a,b] \setminus D_k'\}.$$

Then $\mathcal{C}$ is an open cover of the compact interval $[a,b]$. Let $\lambda > 0$ be the Lebesgue number of the open cover $\mathcal{C}$ of $[a,b]$. Let P be a partition of $[a,b]$ with mesh $\mu(P) < \lambda$. Then each subinterval $[x_{i-1}, x_i]$ of P is either contained in J_j for some $j \in \mathbb{N}$ or contained in I_x for some $x \in [a,b] \setminus D_k'$.

Let $A = \{i : [x_{i-1}, x_i] \subseteq J_j \text{ for some } j\}$ and $B = \{1, 2, \ldots, n\} \setminus A$. Now, we have

$$\begin{aligned}
U(f,P) - L(f,P) &= \sum_{i \in A}(M_i - m_i)\ \Delta x_i + \sum_{i \in B}(M_i - m_i)\ \Delta x_i \\
&\leq (2M) \sum_{i \in A} \Delta x_i + \frac{1}{k}\sum_{i \in B} \Delta x_i \\
&< (2M)\left(\sum_{j=1}^{\infty} (b_j - a_j)\right) + \frac{1}{k}\ (b-a) \\
&< (2M)\left(\frac{\epsilon}{4M}\right) + \frac{\epsilon}{2} = \epsilon.
\end{aligned}$$

By Riemann's integrability criterion, we have $f \in \mathcal{R}[a,b]$.)